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1 July 1975
SUPERSEDING
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6 March 1964

MILITARY SPECIFICATION

ADAPTERS, PALLET, STEEL FOR CYLINDRICAL CONTAINERS

THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

1. SCOPE

1.1 Scope. This specification covers the requirements for the procurement and testing of two types of steel pallet adapters for cylindrical containers.

1.2 Classification. Pallet adapters shall be of the following types, MK's and MOD's as specified (see 6.2).

Type I - Resistance welded construction (6.4)

MK and MOD

MK 12 MOD 1 (for 3"/50 Cartridge Tank)
MK 13 MOD 0 (for 5"/38 Cartridge Tank)
MK 14 MOD 0 (for 5.0 Rocket Containers MK 10 MOD 0 & 1)
MK 15 MOD 0 (for Aero Launcher 6A)
MK 16 MOD 0 (for 5"/54 Cartridge Tank)
MK 17 MOD 0 (for 76mm Cartridge)
MK 18 MOD 0 (for 6"/47 Cartridge Tank, MK 1, MK 3, MK 4 & MODS)
MK 121 MOD 0 (for 76mm Cartridge Tank)

Type II - Metal arc welded construction (6.4)

MK and MOD

MK 19 MOD 0 (for uncrated Depth Charge Case MK 9 & MODS)
MK 22 MOD 0 (for Warhead and/or Exercise Head (TERRIER/TARTAR) in container MK 211 MOD 0)
MK 23 MOD 0 (for AN8024, AN8025, MK 209 MOD 0 & MK 260 MOD 0 Containers)
MK 24 MOD 0 (for MS63052 Style 1 Drum)
MK 25 MOD 0 (for AN8027 & MK 259 MOD 0 Containers)
MK 46 MOD 0 (for 4 Aero 70, 4 LAU-3A/A and 9 LAU-32A/A Launchers)

FSC - 3990

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS:

* Federal

QQ-S-781	Strapping, Steel, Flat and Seals

* Military

MIL-W-6858	Welding, Resistance: Aluminum, Magnesium, Non-Harding Steels or Alloys, Nickel Alloys, Heat Resisting Alloys, and Titanium Alloys; Spot and Seam
MIL-W-21157	Weldment, Steel, Carbon and Low Alloy (Yield Strength 30,000-60,000 PSI)

* STANDARDS:

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-248	Qualification Test for Welders

DRAWINGS AND LISTS:

* Naval Sea Systems Command (Code Ident 10001)

LD 296475	Adapter, Pallet, MK 13 MOD 0
LD 296477	Adapter, Pallet, MK 14 MOD 0
LD 296478	Adapter, Pallet, MK 15 MOD 0
LD 296481	Adapter, Pallet, MK 17 MOD 0
LD 296485	Adapter, Pallet, MK 12 MOD 1
LD 296489	Adapter, Pallet, MK 16 MOD 0
LD 483312	Adapter, Pallet, MK 18 MOD 0
LD 483340	Adapter, Pallet, MK 19 MOD 0
LD 538093	Adapter, Pallet, MK 46 MOD 0
LD 538127	Adapter, Pallet, MK 23 MOD 0
LD 538128	Adapter, Pallet, MK 24 MOD 0
LD 538129	Adapter, Pallet, MK 25 MOD 0
LD 538130	Adapter, Pallet, MK 22 MOD 0
DL 2642761	Adapter, Pallet, MK 121 MOD 0

And all documents referenced in the above listed LD's and DL's. Contractors will require only drawings of specific adapters specified in the contract, requisition or order.

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other Publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

AMERICAN WELDING SOCIETY

AWS A2.0 Welding Symbols

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

3. REQUIREMENTS

3.1 General Requirements.

* 3.1.1 Preproduction Sample. Unless otherwise specified, as soon as practicable after award of contract or order, and prior to submission of any adapter for acceptance, the contractor shall furnish a preproduction sample for determination of conformance to this specification. Examination and tests shall be those specified herein, see 4.2.1. Any changes or deviations from the preproduction sample will be subject to the approval of the contracting officer. The preproduction sample may not be included in the quantity to be shipped in the final shipment.

3.1.2 Construction. Unless otherwise specified in the contract or order, adapters covered by this specification shall be manufactured in strict accordance with the below listed drawings:

* NOMENCLATURE	LIST OF DRAWINGS
Adapter, Pallet, MK 12 MOD 1	LD 296485
Adapter, Pallet, MK 13 MOD 0	LD 296475
Adapter, Pallet, MK 14 MOD 0	LD 296477
Adapter, Pallet, MK 15 MOD 0	LD 296478
Adapter, Pallet, MK 16 MOD 0	LD 296489
Adapter, Pallet, MK 17 MOD 0	LD 296481
Adapter, Pallet, MK 18 MOD 0	LD 483312
Adapter, Pallet, MK 19 MOD 0	LD 483340
Adapter, Pallet, MK 22 MOD 0	LD 538130
Adapter, Pallet, MK 23 MOD 0	LD 538127
Adapter, Pallet, MK 24 MOD 0	LD 538128
Adapter, Pallet, MK 25 MOD 0	LD 538129
Adapter, Pallet, MK 46 MOD 0	LD 538093
Adapter, Pallet, MK 121 MOD 0	DL 2642761

3.2 Welding.

* 3.2.1 Drawing Symbols. Welding symbols on applicable drawings shall be interpreted in accordance with American Welding Society AWS A2.0.

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* 3.2.2 Resistance Welding. Resistance welding (6.3) for Type I adapters shall be performed in accordance with the requirements of MIL-W-6858, Group B. When subjected to the acceptance testing of 4.4.2.2.1 resistance welds shall not fail under the loads specified in applicable drawings.

* NOTE: The strength requirements of these welds for conditions of shear are called-out as a part of the weld symbol, see AWS A2.0; 2.2 or in the general notes of applicable drawings (see 3.2.1).

Resistance welding shall also be capable of meeting the Quality Assurance Provisions of MIL - W - 6858, including the shear test specified therein when subjected to the Production Quality Control testing of 4.4.1.

3.2.2.1 Qualification of Machines. All resistance welding machines
* intended for use during production shall be qualified for the applicable material and particular combination of sheet thickness required by MIL-W-6858.

3.2.3 Metal Arc Welding. Unless otherwise specified in the contract, order, or on applicable drawings, metal arc welding (6.3) for Type II adapters shall be performed in such a manner that it will meet all weldment requirements of MIL-W-21157.

3.2.3.1 Proof of Capability and Welding Operators Qualification. As specified by MIL - W-21157 the contractor shall be required to furnish proof of capability to perform this welding. Welders must be qualified to meet production sample requirements as specified in MIL-STD-248.

3.2.4 Visual Qualifications. When examined as specified in 4.4.2.1, all welding on equipment supplied under this specification shall be smooth and of uniform contour, free from cracks, electrode pick-up, surface burning and metal expulsion.

3.3 Workmanship. All surfaces and edges shall be free from sharp
* corners, rough spots and similar defects that could damage items placed in the adapter or create a hazard to operating personnel. All bends of a major character shall be made with dies or fixtures to insure uniformity of size and shape. Frames shall be free from distortion that will affect assembly of the unit load. The adapters shall be constructed and finished in a manner to assure compliance with this specification.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of Tests. The examination and testing of Pallet Adapters shall be classified as:

(a) Preproduction tests.

(b) Production quality control tests.

(c) Lot acceptance tests.

4.2.1 Preproduction Tests. Examination and testing of the preproduction sample shall be made for all applicable provisions of this specification. These examinations and tests shall be performed in the presence of a Government inspector prior to any acceptance by the Government and need not be repeated during the life of the contract unless a change is made in design, material, or methods of construction.

4.2.2 Production Quality Control Tests for Resistance Welding. Specimens for production quality control tests shall be subjected to the tests of 4.4.1.

4.2.3 Lot Acceptance Tests. These tests shall be accomplished on pallet adapters of each type being submitted to the Government for acceptance. Inspection and test procedures shall be submitted for Government approval prior to the commencement of production, 4.4.2.

4.3 Sampling.

* 4.3.1 Resistance Welding for Production Quality Control. Sampling for production quality control inspection, of resistance welding only, shall consist of three single spot shear test specimens, fabricated, taken and tested as specified in accordance with MIL-W-6858, Group B.

* 4.3.2 Sampling for Lot Acceptance. The following lot acceptance tests shall be performed in accordance with MIL-STD-105:

(a) Visual and dimensional examination (4.4.2.1) - Inspection Level II Acceptable Quality Levels (AQL's) 1.0% for major defects and 2.5% for minor defects. To identify major, minor, or critical defects, see drawings listed in 3.1.2. In regard to critical defects class, for inspection lot sizes of 150 or less, use 100% of the lot. For inspection lot sizes of 151-3200, use sample of 150 units, acceptance number zero. For inspection lot sizes greater than 3200, use the sample size as indicated in MIL-STD-105, acceptance number zero.

(b) Protective coating (4.4.2.3) - Inspection Level S3, Acceptable Quality Level (AQL) 1.0%.

(c) Preservation, packaging and packing (4.4.3) - Inspection Level S-2, acceptance number zero.

4.4 Inspection.

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4.4.1 Production quality control testing for resistance welding.

4.4.1.1 Examination. Specimens selected in accordance with 4.3.1 shall be visually examined for compliance with the requirements of 3.2.4.

4.4.1.2 Testing. Specimens selected in accordance with 4.3.1 shall be tested to determine compliance with the requirements of MIL-W-6858, Group B, including the shear test specified therein.

4.4.2 Lot acceptance tests.

* 4.4.2.1 Visual and Dimensional Examination. Each pallet adapter selected in accordance with 4.3.2 shall be visually examined for completeness of manufacture, freedom of defects as described in 3.2.4., proper identification, and workmanship. Dimensional and other inspections shall be performed in accordance with listed drawings (see 3.1.2).

4.4.2.2 Testing.

4.4.2.2.1 Resistance Welding. Each production lot shall be certified by shear testing in accordance with 4.3.1. Each shear test specimen (6.3) shall be loaded so as to place the weld in shear. Welds shall not fail when the magnitude of the shear loading is equal to or exceeds that specified on applicable drawings or specification (see 3.2.2).

4.4.2.2.2 Metal Arc Welding. Each pallet adapter containing arc (6.3) welded components and which has been selected in accordance with 4.3.2, shall be subjected to the applicable testing of MIL-W-21157, class as specified on the applicable drawings.

4.4.2.3 Protective Coating. The pallet adapter shall be examined to determine conformance to 3.1.2.

4.4.3 Preservation, Packaging and Packing. Packs of pallet adapters selected in accordance with 4.3.2 shall be examined to determine conformance to Section 5 of this specification.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and Packaging. Preservation and packing shall be level C.

5.1.1 Unit Packaging, Level C. Adapter sets having more than one component shall have component parts nested and each set secured with two or more straps, QQ-S-781, Type I, Class B, or galvanized wire binders. Individual adapters shall be properly nested.

5.2 Packing. Not applicable.

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☆ 5.3 Palletizing. Components unitized in accordance with 5.1.3.1 shall be stacked on wood pallets to a height not exceeding 72 inches. Each pallet shall have wood or metal battens and shall be secured with circumferential steel strapping so as to provide a firm load. A minimum of two additional straps, lengthwise and widthwise, QQ-S-781, Type I, Class B, shall be used to secure the load to the pallet.

5.4 Marking. In addition to any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

6 NOTES

6.1 Intended Use. The pallet adapters covered by this specification are intended for use in conjunction with a pallet and steel strapping to form a palletized unit load. These devices are required to maintain the position of cylindrical items or containers on a pallet, and to provide a stable base for tiered loads.

* 6.2 Ordering data. DD Form 1423 and other related procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) Type and MK and MOD number of adapter being procured (see 3.1.2).

(c) Whether a preproduction sample is required and, if so, how many (see 3.1.1).

(d) Welding specifications, if different than herein required (see 3.2.2 and 3.2.3).

(e) Minimum lot size, if applicable.

(f) Address of activity to receive shipments of acceptable Adapters.

* 6.2.2 Contract data requirements. The items of deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data requirement	Applicable DID
3.2.3.1	Certification, Welder Qualification	*

The DID (Data Item Descriptions/DD Forms 1664) for the above data requirement will be (*) documented in the Department of Defense Authorized Data List (TD-3). Such data will be delivered as identified on the completed DID when specified on DD Form 1423 (Contract Data Requirements Lists).

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6.3 Definitions. For this document the following definitions shall apply:

* (a) Shear Test Specimen - A resistance (or spot) welding sample taken to determine the ultimate shear strength of the weld.

(b) Electrode - As applied to resistance welding, the parts of a welding machine through which the welding currents and, usually, pressures are applied directly to the work.

(c) Metal arc welding - An arc-welding process wherein coalescence is produced by heating with an electric arc between a covered metal electrode and the work. Shielding is obtained from decomposition of the electrode covering or inert gas. Pressure is not used and filler metal is obtained from the electrode or filler wire.

* 6.4 Welded Construction. Although the welded constructon (1.2) is resistance and arc welding some adapters do include both types of welds.

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE

MIL-A-21618B(OS) Adapters, Pallet, Steel For Cylindrical Containers

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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